

CASESTUDY-STUDENT_MANAGEMENT

TABLES CREATED

```
mysql> CREATE TABLE courses (
->   course_id INT AUTO_INCREMENT PRIMARY KEY,
->   course_name VARCHAR(100),
->   course_description TEXT,
->   credits INT
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE enrollments (
->   enrollment_id INT AUTO_INCREMENT PRIMARY KEY,
->   student_id INT,
->   course_id INT,
->   enrollment_date DATE,
->   grade VARCHAR(2),
->   FOREIGN KEY (student_id) REFERENCES students(student_id),
->   FOREIGN KEY (course_id) REFERENCES courses(course_id)
-> );
Query OK, 0 rows affected (0.09 sec)

mysql> show tables;
+-----+
| Tables_in_studentmanagement |
+-----+
| courses                      |
| enrollments                  |
| students                     |
+-----+
3 rows in set (0.04 sec)

mysql>
```

INSERTED DATA INTO THE TABLES

```
mysql> select * from students;
+-----+-----+-----+-----+-----+-----+
| student_id | first_name | last_name | dob       | email                               | phone       |
+-----+-----+-----+-----+-----+-----+
| 1          | John      | Doe       | 2002-05-14 | john.doe@example.com              | 9876543210 |
| 2          | Jane      | Smith     | 2003-08-21 | jane.smith@example.com            | 1112223333 |
| 3          | Peter     | Jones     | 2001-02-10 | peter.jones@example.com           | 4445556666 |
| 4          | Susan     | Davis     | 2004-11-05 | susan.davis@example.com           | 7778889999 |
| 5          | Hema      | Sree      | 2004-11-05 | hema.sree@example.com             | 7778889999 |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select * from courses;
+-----+-----+-----+-----+
| course_id | course_name          | course_description                                     | credits |
+-----+-----+-----+-----+
| 1          | Web Development      | Learn HTML, CSS, JavaScript                           | NULL    |
| 2          | Database Management  | Introduction to SQL and database design                 | NULL    |
| 3          | Data Structures      | Study of fundamental data structures and algorithms     | NULL    |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> select * from enrollments;
+-----+-----+-----+-----+-----+
| enrollment_id | student_id | course_id | enrollment_date | grade |
+-----+-----+-----+-----+-----+
| 6             | 1          | 1          | 2025-08-05      | NULL  |
| 7             | 2          | 2          | 2025-08-05      | NULL  |
| 8             | 3          | 1          | 2025-08-05      | NULL  |
| 9             | 4          | 3          | 2025-08-05      | NULL  |
| 10            | 5          | 2          | 2025-08-05      | NULL  |
+-----+-----+-----+-----+-----+
```

CASESTUDY-STUDENT_MANAGEMENT

TASK1: Add 3 new students with different details.

```
mysql> INSERT INTO students (first_name, last_name, dob, email, phone)
-> VALUES
-> ('Alice', 'Smith', '2003-08-21', 'alice.smith@example.com', '9123456780'),
-> ('Bob', 'Johnson', '2001-11-02', 'bob.johnson@example.com', '9234567890'),
-> ('Charlie', 'Brown', '2002-03-15', 'charlie.brown@example.com', '9345678901');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from students;
```

student_id	first_name	last_name	dob	email	phone
1	John	Doe	2002-05-14	john.doe@example.com	9876543210
2	Jane	Smith	2003-08-21	jane.smith@example.com	1112223333
3	Peter	Jones	2001-02-10	peter.jones@example.com	4445556666
4	Susan	Davis	2004-11-05	susan.davis@example.com	7778889999
5	Hema	Sree	2004-11-05	hema.sree@example.com	7778889999
6	Alice	Smith	2003-08-21	alice.smith@example.com	9123456780
7	Bob	Johnson	2001-11-02	bob.johnson@example.com	9234567890
8	Charlie	Brown	2002-03-15	charlie.brown@example.com	9345678901

```
8 rows in set (0.00 sec)
```

TASK2: Create 2 new courses.

```
mysql> INSERT INTO courses (course_name, course_description)
-> VALUES
-> ('Data Structures', 'Learn about arrays, linked lists, trees, and graphs'),
-> ('Database Systems', 'Introduction to relational databases and SQL');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> select * from courses;
```

course_id	course_name	course_description	credits
1	Web Development	Learn HTML, CSS, JavaScript	NULL
2	Database Management	Introduction to SQL and database design	NULL
3	Data Structures	Study of fundamental data structures and algorithms	NULL
4	Data Structures	Learn about arrays, linked lists, trees, and graphs	NULL
5	Database Systems	Introduction to relational databases and SQL	NULL

```
5 rows in set (0.00 sec)
```

Task3: Enroll each student in at least one course.

```
mysql> INSERT INTO enrollments (student_id, course_id, enrollment_date, grade)
-> VALUES (2, 1, CURDATE(), NULL);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO enrollments (student_id, course_id, enrollment_date, grade)
-> VALUES (3, 2, CURDATE(), NULL);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO enrollments (student_id, course_id, enrollment_date, grade)
-> VALUES (4, 3, CURDATE(), NULL);
Query OK, 1 row affected (0.00 sec)

mysql> select * from enrollments;
```

enrollment_id	student_id	course_id	enrollment_date	grade
6	1	1	2025-08-05	NULL
7	2	2	2025-08-05	NULL
8	3	1	2025-08-05	NULL
9	4	3	2025-08-05	NULL
10	5	2	2025-08-05	NULL
11	2	1	2025-08-05	NULL
12	3	2	2025-08-05	NULL
13	4	3	2025-08-05	NULL

```
13 rows in set (0.00 sec)
```

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TASK4: Update one student's name and email.

```
mysql> select * from students;
```

student_id	first_name	last_name	dob	email	phone
1	John	Doe	2002-05-14	john.doe@example.com	9876543210
2	Jane	Smith	2003-08-21	jane.smith@example.com	1112223333
3	Peter	Jones	2001-02-10	peter.jones@example.com	4445556666
4	Charles	Davis	2004-11-05	charles.brown@example.com	7778889999
5	Hema	Sree	2004-11-05	hema.sree@example.com	7778889999
6	Alice	Smith	2003-08-21	alice.smith@example.com	9123456780
7	Bob	Johnson	2001-11-02	bob.johnson@example.com	9234567890
8	Charlie	Brown	2002-03-15	charlie.brown@example.com	9345678901

TASK5: Change grades for 2 enrollments.

```
mysql> UPDATE enrollments SET grade = 'A' WHERE enrollment_id = 3;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> select * from enrollments;
```

enrollment_id	student_id	course_id	enrollment_date	grade
6	1	1	2025-08-05	NULL
7	2	2	2025-08-05	NULL
8	3	1	2025-08-05	NULL
9	4	3	2025-08-05	NULL
10	5	2	2025-08-05	NULL
11	2	1	2025-08-05	NULL
12	3	2	2025-08-05	NULL
13	4	3	2025-08-05	NULL

```
8 rows in set (0.00 sec)
```

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TASK6: Write a query to list all students enrolled in “Web Development”.

```
mysql> SELECT s.first_name, s.last_name, c.course_name
-> FROM enrollments e
-> JOIN students s ON e.student_id = s.student_id
-> JOIN courses c ON e.course_id = c.course_id
-> WHERE c.course_name = 'Web Development';
```

first_name	last_name	course_name
John	Doe	Web Development
Peter	Jones	Web Development
Jane	Smith	Web Development

3 rows in set (0.00 sec)

TASK7: Delete one course and ensure it doesn't break other relationships.

```
mysql> DELETE FROM courses WHERE course_id = 2;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> select * from courses;
```

course_id	course_name	course_description	credits
1	Web Development	Learn HTML, CSS, JavaScript	NULL
3	Data Structures	Study of fundamental data structures and algorithms	NULL
4	Data Structures	Learn about arrays, linked lists, trees, and graphs	NULL
5	Database Systems	Introduction to relational databases and SQL	NULL

4 rows in set (0.01 sec)